Introduction

Trim-and-fill, introduced by Duval and Tweedie in 2000, is the most widely used algorithmic approach for adjusting a pooled meta-analytic estimate when the funnel plot looks asymmetric. The idea is intuitive: if some smaller studies on the side opposite the dominant tail of the funnel appear to be missing — possibly because they were never published, never indexed, or never reached the systematic-review screening stage — the procedure imputes mirror-image counterparts and re-pools the augmented set. The difference between the unadjusted and adjusted estimates quantifies how sensitive the conclusion is to the presence of these hypothetical missing studies. Crucially, trim-and-fill is a sensitivity analysis, not a method of bias correction in the strict statistical sense; it cannot recover the true unbiased effect.

Prerequisites

Familiarity with funnel plots, the small-study-effect concept, the inverse-variance random-effects pool, and the distinction between sensitivity analysis and bias correction.

Theory

The algorithm proceeds iteratively:

1. Estimate the number of asymmetric studies in the dominant tail using one of three rank-based estimators (L_0, R_0, Q_0).
2. Trim those studies from the dominant side and re-estimate the pooled effect; this estimate becomes the new symmetry centre.
3. Fill the analysis by reflecting the trimmed studies across the new centre to the opposite side.
4. Re-pool with the augmented dataset to obtain the adjusted estimate.

Iterating to convergence yields the final imputed studies and adjusted pooled estimate. The number of imputed studies is itself an interpretable summary: zero imputations indicate no detectable asymmetry, and a large number indicates substantial sensitivity.

Assumptions

The observed funnel asymmetry is attributable to selection or publication bias rather than to true between-study heterogeneity correlated with study size; the underlying effect distribution is mirror-symmetric around the true effect; and the pooled model is otherwise correctly specified.

R Implementation

```r
library(metafor)

set.seed(2026)
k <- 18
yi <- rnorm(k, 0.35, 0.25)
vi <- 1 / sample(15:300, k, replace = TRUE)

keep <- !(sqrt(vi) > 0.1 & abs(yi) < 0.15)
yi_b <- yi[keep]; vi_b <- vi[keep]

res <- rma(yi_b, vi_b, method = "REML")
tf  <- trimfill(res)

data.frame(method = c("Original", "Trim-and-fill"),
           estimate = round(c(res$b, tf$b), 3),
           ci = c(sprintf("[%.2f, %.2f]", res$ci.lb, res$ci.ub),
```

```r
                  sprintf("[%.2f, %.2f]", tf$ci.lb, tf$ci.ub)))

funnel(tf, main = "Trim-and-fill adjusted funnel")
```

Output & Results

`trimfill()` returns the number of imputed studies, the adjusted pooled estimate with its confidence interval, and an augmented dataset suitable for plotting on a funnel diagram. Reporting both the original pool and the adjusted pool side by side — together with the count of imputations — lets readers see exactly how much the conclusion shifts when the missing studies are introduced.

Interpretation

A reporting sentence: "Trim-and-fill (Duval and Tweedie, L_0 estimator) imputed four potentially missing studies on the left side of the funnel; the adjusted standardised mean difference was 0.18 (95 % CI 0.03 to 0.33), notably attenuated from the unadjusted estimate of 0.32 (0.21 to 0.43). The result suggests the pooled conclusion is sensitive to plausible publication bias, and Egger's regression and PET-PEESE were reported as triangulating sensitivity analyses." Always report the unadjusted estimate as primary and the adjusted estimate as sensitivity.

Practical Tips

- Trim-and-fill is a sensitivity analysis, not a proof of publication bias; report it as such, and never use the adjusted estimate as the primary effect.
- The method performs poorly when funnel asymmetry is driven by genuine between-study heterogeneity correlated with study size; in such settings it can spuriously imply bias.
- Apply alongside Egger's regression test and PET-PEESE for triangulation; consistent signals strengthen the bias interpretation, divergent signals warn against over-interpretation.
- Report both original and adjusted estimates with their CIs and the count of imputed studies; an imputation count of zero indicates the funnel was already symmetric and no adjustment was needed.
- The L_0 estimator (default in `metafor`) is the most widely used; R_0 is more conservative; Q_0 is rarely used in practice. Sensitivity to estimator choice should be noted when results are borderline.
- Selection-model approaches (such as Vevea–Hedges weight-function models) and PET-PEESE are increasingly preferred over trim-and-fill in methodological literature; consider them when the small-study effect is consequential to your conclusion.

R Packages Used

`metafor::trimfill()` for the canonical implementation with all three estimators and `funnel()` overlay; `meta::trimfill()` for the alternative interface within the `meta` ecosystem; `metasens` for trim-and-fill alongside copas and limit-meta-analysis bias-adjustment methods; `weightr` and `puniform` for selection-model alternatives that handle complex bias mechanisms.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis